THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The pair of equations $3x + 2y = 5$ and $9x + 6y = 15$ is: 1

(a) infinitely many solutions (coincident lines)
(b) exactly two solutions
(c) exactly one solution (intersecting lines)
(d) no solution (parallel lines)

2. The solution of $x + y = 8$, $x - y = 2$ has $x =$ 1

(a) 4 **(b)** 7 **(c)** 6 **(d)** 5

3. The solution of $2x + 3y = 12$, $5x + 4y = 23$ has $y =$ 1

(a) 4 **(b)** 1 **(c)** 2 **(d)** 3

4. For the pair $x + y = 10$ and $x - y = 2$, the value of xy is: 1

(a) 26 **(b)** 25 **(c)** 23 **(d)** 24

5. The pair $x + 2y = 3$ and $2x + ky = 7$ has no solution when $k =$ 1

(a) 5 **(b)** 6 **(c)** 3 **(d)** 4

6. The point where the lines $7x + 2y = -6$ and $3x + 5y = 14$ intersect has $x + y =$ 1

(a) 2 **(b)** 1 **(c)** 4 **(d)** 3

7. The pair of equations $2x + 3y = 8$ and $4x + 6y = 12$ is: 1

(a) exactly one solution (intersecting lines)
(b) exactly two solutions
(c) infinitely many solutions (coincident lines)
(d) no solution (parallel lines)

8. The solution of $2x + y = 10$, $x - y = 2$ has $x =$ 1

(a) 6 **(b)** 3 **(c)** 5 **(d)** 4

9. The solution of $3x + 2y = 14$, $2x - 3y = 5$ has $y =$ 1

(a) 2 **(b)** 0 **(c)** 1 **(d)** 3

10. For the pair $x + y = 14$ and $x - y = 4$, the value of xy is: 1

(a) 46 **(b)** 45 **(c)** 47 **(d)** 44

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A man tells his daughter, "Seven years ago I was 7 times as old as you were. Three years from now I shall be 3 times as old as you will be." Form the pair of linear equations (do not solve). **2**

12. Solve by substitution: $x + y = 8$, $x - y = 2$. **2**

13. Find the value of k for which the pair $x + 2y = 3$, $2x + ky = 7$ has no solution. **2**

14. A man tells his daughter, "Seven years ago I was 6 times as old as you were. Three years from now I shall be 3 times as old as you will be." Form the pair of linear equations (do not solve). **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. On comparing the ratios, find whether the lines $5x - 4y = 8$ and $7x + 6y = 9$ intersect, are parallel or coincide. **3**

16. Solve by elimination method: $2x + 3y = 12$ and $5x + 4y = 23$. **3**

17. The sum of the digits of a two-digit number is 9. If the digits are interchanged, the number becomes 45 less. Find the number. **3**

OR

On comparing the ratios, find whether the lines $9x + 3y = 12$ and $18x + 6y = 24$ intersect, are parallel or coincide. **3**

18. Solve by elimination method: $3x + 2y = 14$ and $2x - 3y = 5$. **3**

OR

The sum of the digits of a two-digit number is 9. If the digits are interchanged, the number becomes 27 less. Find the number. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A father is 8 times as old as his son. After 9 years the father will be 5 times as old as the son will be. Find their present ages. **5**

OR

The taxi charges consist of a fixed charge plus a charge per km. For 10 km the fare is ₹170 and for 15 km it is ₹240. Find the fixed charge, the charge per km and the fare for 25 km. **5**

20. A father is 7 times as old as his son. After 6 years the father will be 4 times as old as the son will be. Find their present ages. **5**

OR

The taxi charges consist of a fixed charge plus a charge per km. For 8 km the fare is ₹141 and for 14 km it is ₹213. Find the fixed charge, the charge per km and the fare for 30 km. **5**

END OF QUESTION PAPER — Total: 40 Marks